

2008

Time Allowed: 3 hours plus 5 minutes reading time

- Write your name and class at the top of this page, and at the top of each answer sheet
- At the end of the examination this examination paper must be attached to the front of your answers
- All questions are of equal value and may be attempted
- All necessary working must be shown. Marks may not be awarded for careless or badly arranged work.
- Marks indicated are a guide only and may be varied if necessary.

(for Markers Use Only)

[illegible]

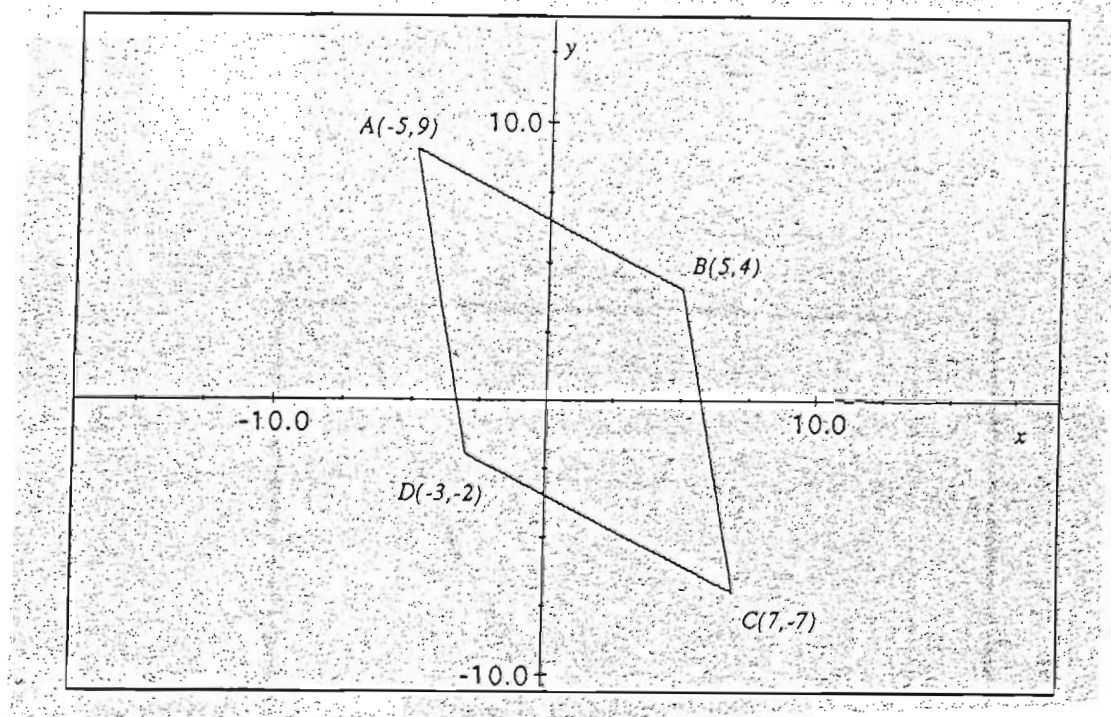
Question 1 (12 marks)

- a) Find $e^{-0.6}$ correct to 3 decimal places. 1
- b) Expand and simplify $(\sqrt{2}-3)^2$ 2
- c) Given $\frac{1}{P} = \frac{1}{Q} + \frac{1}{R}$ make Q the subject of the formula. 2
- d) (i) Find $\int_1^2 \frac{dx}{x}$ 1
- (ii) Evaluate $\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \cos\left(\frac{x}{2}\right) dx$. Leave your answer as an exact value. 2
- e) Solve the inequality $|2x - 3| \leq 7$ 2
- f) Solve the following equations simultaneously
- $2x + y = 4$
- $5x + 2y = 9$ 2

Question 2 (Use a separate sheet of paper) (12 marks)

- a) A rhombus is a parallelogram with four sides of equal length.

The figure shown below, with vertices $A(-5, 9)$, $B(5, 4)$, $C(7, -7)$ and $D(-3, -2)$ is a rhombus.

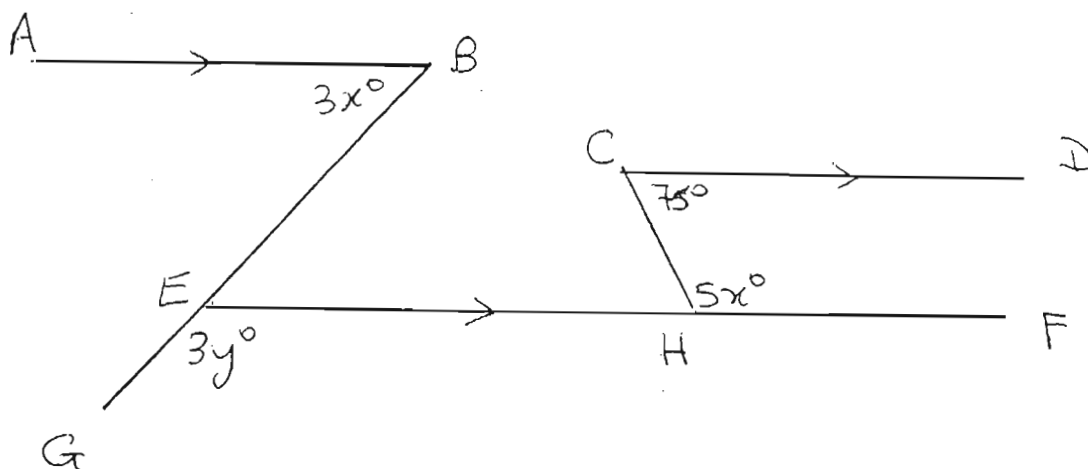


- | | | |
|-------|--|---|
| (i) | Find the side length of $ABCD$. Give your answer in simplified surd form. | 1 |
| (ii) | Find the gradient of the longer diagonal. | 1 |
| (iii) | Show that the diagonals of $ABCD$ are perpendicular. | 2 |
| (iv) | Find the coordinates of the midpoint of each diagonal. | 1 |
| (v) | What does this result to part (d) say about the diagonals of this rhombus? | 1 |
| (vi) | Find the equation of the line passing through AC . | 2 |

(b) In the diagram below the lines AB, CD and EF are parallel.

4

Find the value of x and y . Give reasons for each answer.

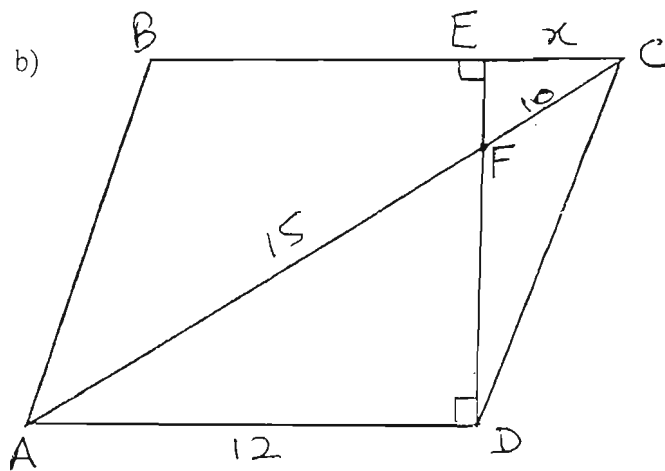


Question 3 (12 marks) (Use a separate sheet of paper)

- a) Differentiate
- (i) $x^2 e^x$ 2
- (ii) $\ln\left(\frac{x-5}{x+3}\right)$ 2
- b) (i) Find $\int \frac{dx}{3x-1}$ 1
- (ii) Evaluate $\int_0^1 e^{4x} dx$, leaving your answer in exact form 2
- c) For what values of m does the equation $4x^2 + (1+m)x + 1 = 0$ have equal roots. 2
- d) For acute angles A and B it is given that $\sin A = \frac{12}{13}$ and $\cos B = \frac{15}{17}$
- Find the exact value of $\sec A + \tan B$. 3

Question 4 (12 marks) (Use a separate sheet of paper)

- a) The sum of the first 4 terms of a geometric progression is 30, and the limiting sum is 32. If the common ratio is negative find the first three terms. 3



$ABCD$ is a parallelogram.

(i) Prove that $\triangle EFC$ and $\triangle DFA$ are similar.

(ii) Find the value of x .

4

Not to Scale

c) Solve $\sin\left(x + \frac{\pi}{3}\right) = 0$ for $0 \leq x \leq \pi$

2

d) α and β are the roots of $2x^2 - 5x + 5 = 0$. Write down the value of

(i) $\alpha + \beta$

(ii) $\alpha \beta$

(iii) $\frac{1}{\alpha} + \frac{1}{\beta}$

3

Question 5 (12 marks) (Use a separate sheet of paper)

a) A function is defined by $f(x) = 3x^2 - 2x^3$

(i) Find the coordinates of any turning points and determine their nature

3

(ii) Sketch the curve, indicating all intercepts and turning points.

2

(iii) State the domain over which both $f(x) > 0$ and $f'(x) > 0$

1

(iv) On the same set of axes sketch the line $f(x) = \frac{1}{2}$

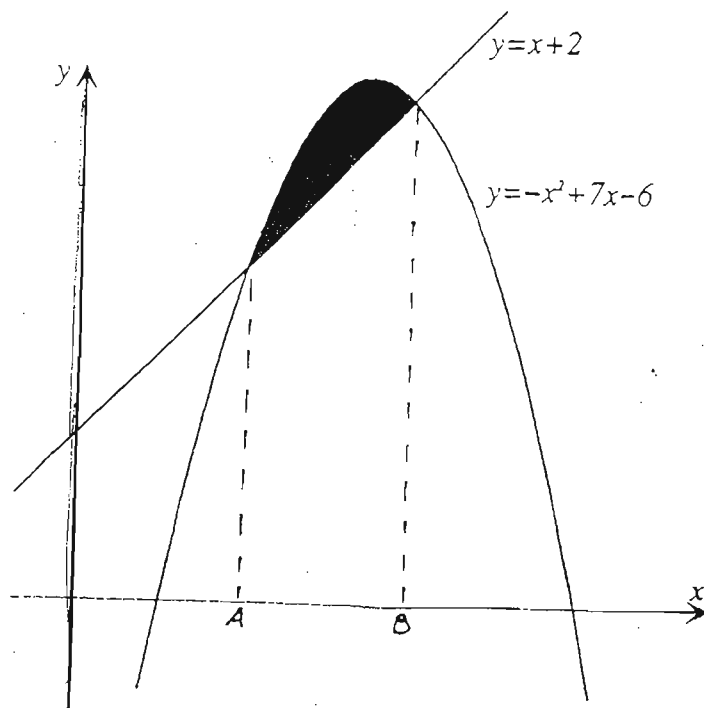
1

(v) Hence find the number of solutions to the equation $6x^2 - 4x^3 = 1$

1

b)

4



The diagram shows the graphs of the functions $y = -x^2 + 7x - 6$ and $y = x + 2$.

- (i) Show that the value of A and B is 2 and 4 respectively
- (ii) Calculate the area of the shaded region.

Question 6 (12 marks) (Use a separate sheet of paper)

a) Evaluate $\sum_{r=1}^4 3^{1-r}$ 1

b) For the arithmetic progression 32, 25, 18,

- | | | | |
|----------|-------|-------------------------------------|---|
| find the | (i) | the 15 th term | 1 |
| | (ii) | S_{15} | 1 |
| | (iii) | the sum of the <u>next</u> 20 terms | 2 |

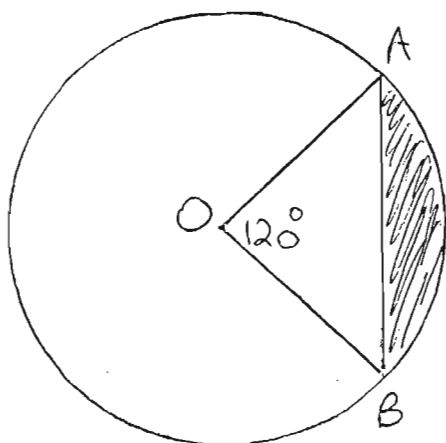
- c) The area under the curve $y = 4^x$ between $x = 0$ and $x = 2$ is rotated about the x -axis. Copy and complete the table.

x	0	0.5	1	1.5	2
4^{2x}					

Use your results with Simpson's rule to find an approximate value for the volume of revolution. Use 5 function values and answer correct to 1 decimal place.

3

d)



The circle has a radius of 2cm

- (i) Find arc length AB
(ii) Find the shaded area
(correct to 1 decimal place)

4

Question 7 (12 marks) (Use a separate sheet of paper)

a) $f'(x) = 3x^2 - 4$.

Find $y = f(x)$ if the function passes through (3, 8).

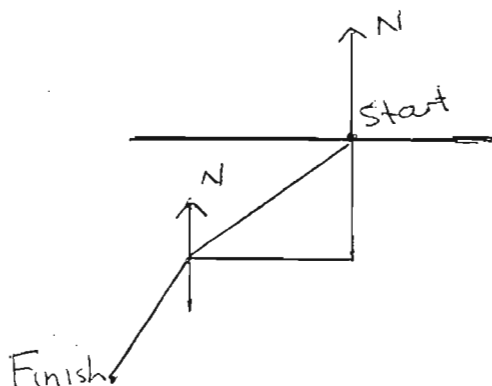
2

- b) A boat travels 5km on a bearing of 207° T, then travels 8km on a bearing of 200° T.

Find the straight line distance between the start and finish to 3 significant figures.

4

Copy and complete the given diagram to assist your working.



- c) \$30 000 is borrowed to buy a car. Interest is charged at 12% pa, compounding monthly.

The loan is repaid in equal monthly repayments over 4 years. Let A_n be the amount owing after n months.

- (i) If M is the monthly payment write an expression for the amount owing

after α) 1 month

β) 3 months

- (ii) Find M

- (iii) Find the total amount paid over the 4 years.

6

Question 8 (12 marks) (Use a separate sheet of paper)

- a) Evaluate $\lim_{x \rightarrow 0} \frac{\sin 2x}{x}$ 2

- b) Evaluate $\log_5 100 - \log_5 4$ 2

- c) A particle moves in such a way that its distance, x metres, from the origin after t seconds is given by

$$x = 2 + 3t - t^3 \text{ for } t > 0$$

- (i) Find an equation for its velocity after t seconds. 1
- (ii) At what time does the particle stop? 1
- (iii) Where is the particle initially? 1
- (iv) Find the velocity after 2 seconds. 1
- (v) How far has the particle travelled in the first 2 seconds. 2
- d) Find the volume of the solid formed when the curve $y = \sqrt{x}$ is rotated about the x axis between $x = 1$ and $x = 5$. (leave the answer in terms of π). 2

Question 9 (12 marks) (Use a separate sheet of paper)

a) If $F(x) = \begin{cases} x^2 - 2 & x \leq -1 \\ 2^x & -1 < x < 2 \\ \log_{10} x & x \geq 2 \end{cases}$

evaluate $f(-1) + f(1) + f(10)$.

2

b) Draw a neat sketch of $y = 3\sin 2x$ within the domain $0 \leq x \leq 2\pi$.

State the (i) period

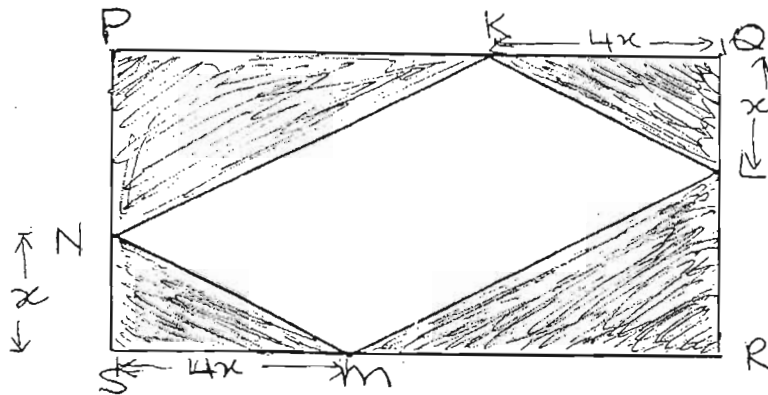
(ii) amplitude.

4

c) In the diagram, PQRS is a rectangle with PQ=40cm, SP=10cm.

The shaded portions are cut away, leaving the parallelogram KLMN.

QL=SN=x cm and QK=SM=4x cm.



(i) Show that the area of the parallelogram KLMN is given by

$$A = 80x - 8x^2$$

3

(ii) Find the allowable values of x

1

(iii) Find the value of x for which A is a maximum

2

Question 10 (12 marks) (Use a separate sheet of paper)

- a) For all values of x in the domain of $0 \leq x \leq 6$, a function $f(x)$ satisfies

$$f'(x) > 0 \text{ and } f''(x) > 0.$$

Sketch a possible graph of $y = f(x)$ in this domain.

2

- b) (i) Find the points of intersection of the curve $y = 4 - \sqrt{2x}$ with the x and y axes. 2

(ii) The area enclosed by the curve $y = 4 - \sqrt{2x}$, the x axis and the y axis is rotated about the y axis. Find the volume of the solid of revolution so formed

(leave your answer in terms of π)

4

- c) The line $x = m$, cuts the curves $y = \log_e x$ and $y = \log_e 5x$ at R and S respectively.

Show that the tangents to the curves at R and S are parallel. Also show that the distance

RS remains constant for all values of M (ie the distance is independent of m).

4

END OF PAPER

Mathematics 2008
HSC Trial Exam

Question 1

a) $e^{-0.6} = 0.549$ (3 dp)

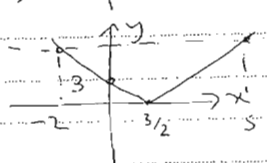
b) $(\sqrt{2} - 3)^2 = 2 - 6\sqrt{2} + 9$
 $= 11 - 6\sqrt{2}$

c) $\frac{1}{P} = \frac{1}{Q} + \frac{1}{R}$
 $\therefore \frac{1}{Q} = \frac{1}{P} - \frac{1}{R}$
 $= \frac{R-P}{PR}$
 $\therefore Q = \frac{PR}{R-P}$

d) (i) $\int_1^2 \frac{dx}{x} = [\ln x]_1^2$
 $= \ln 2 - \ln 1$
 $= \ln 2$

(ii) $\int_{\pi/3}^{\pi/2} \cos\left(\frac{x}{2}\right) dx = 2 \left[\sin\left(\frac{x}{2}\right) \right]_{\pi/3}^{\pi/2}$
 $= 2 \left[\sin\left(\frac{\pi}{4}\right) - \sin\left(\frac{\pi}{6}\right) \right]$
 $= 2 \left[\frac{1}{\sqrt{2}} - \frac{1}{2} \right]$
 $= 2 \left[\frac{\sqrt{2}}{2} - \frac{1}{2} \right]$
 $= \sqrt{2} - 1$

e) $|2x-3| \leq 7$



$2x-3=7$

$2x=10$

$x=5$

$2x-3=-7$

$2x=-4$

$x=-2$

$\therefore -2 \leq x \leq 5$

f) $2x+y=4$ (1)
 $5x+2y=9$ (2)

(1) $\times 5$ $10x+5y=20$ (3)

(2) $\times 2$ $10x+4y=18$ (4)

(3) - (4)

$y=2$

ln (1) $2x+2=4$

$2x=2$

$x=1$

Question 2

a) (i) Using A and B

$$\text{side length} = \sqrt{(-5-5)^2 + (9-4)^2}$$

$$= \sqrt{(10)^2 + (5)^2}$$

$$= \sqrt{125}$$

$$= 5\sqrt{5} \text{ units}$$

(ii) longer diagonal is AC

$$\text{gradient AC} = \frac{9-7}{-5-7}$$

$$= \frac{16}{-12}$$

$$= -4/3 = m_1$$

(iii) shorter diagonal is DB

$$\text{gradient DB} = \frac{-2-4}{-3-5}$$

$$= \frac{-6}{-8}$$

$$= 3/4 = m_2$$

$$\text{Now } m_1 m_2 = \frac{-4}{3} \times \frac{3}{4}$$

$$= -1$$

$\therefore$ Satisfies condition for perpendicular lines

$\therefore$ diagonals perpendicular

(iv)

$$M_{AB} = \left(\frac{-5+7}{2}, \frac{9+7}{2} \right)$$

$$= (1, 1)$$

$$M_{BD} = \left(\frac{-3+5}{2}, \frac{-2+4}{2} \right)$$

$$= (1, 1)$$

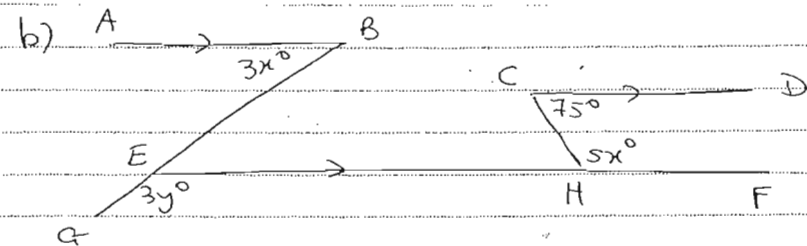
(v) Result confirms diagonals bisect, at (1,1)

(vi) Gradient AC = $-4/3$

$$\therefore \text{Eqn AC : } y-9 = -4/3(x+5)$$

$$3y-27 = -4x-20$$

$$4x+3y-7=0$$



Since $CD \parallel HF$ $75^\circ + 5x^\circ = 180^\circ$

i.e. co-interior angles supplementary

$$\therefore 5x^\circ = 105^\circ$$

$$x = 21$$

Since $AB \parallel EH$, $\angle BEH = 3x^\circ$ (alternate angles equal)

Then $3x + 3y = 180$ (straight angle is 180°)

But $x = 21$

$$\therefore 3y = 180 - 63$$

$$= 117$$

$$y = 39$$

Question 3

a) (i) $y = x^2 e^x$

$u = x^2 \quad v = e^x$

$$y' = x^2 e^x + 2x(e^x) \\ = x e^x (x+2)$$

$u' = 2x \quad v' = e^x$

(ii) $y = \ln\left(\frac{x-5}{x+3}\right)$

$$= \ln(x-5) - \ln(x+3)$$

$$y' = \frac{1}{x-5} - \frac{1}{x+3}$$

$$= \frac{x+3 - (x-5)}{(x-5)(x+3)}$$

$$= \frac{8}{(x-5)(x+3)}$$

$$= \frac{8}{(x-5)(x+3)}$$

b) (i) $\int \frac{dx}{3x-1} = \frac{1}{3} \ln(3x-1) + c$

(ii) $\int_0^1 e^{4x} dx = \left[\frac{1}{4} e^{4x} \right]_0^1$

$$= \frac{1}{4} e^4 - \frac{1}{4} e^0$$

$$= \frac{1}{4} e^4 - \frac{1}{4} = \frac{1}{4} (e^4 - 1)$$

c) $4x^2 + (1+m)x + 1 = 0$

Equal roots when $\Delta = 0$

$$\Delta = b^2 - 4ac$$

$$= (1+m)^2 - 4(4)(1)$$

$$= 1 + 2m + m^2 - 16$$

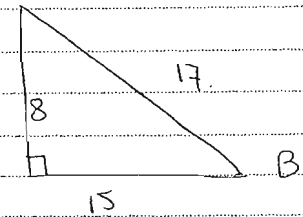
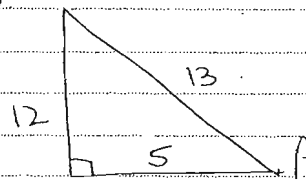
$$= m^2 + 2m - 15$$

Solve $m^2 + 2m - 15 = 0$

$$(m+5)(m-3) = 0$$

$$m = -5 \text{ or } m = 3$$

d)



Complete each triangle.

$$\sec A + \tan B = \frac{13}{5} + \frac{8}{15} \\ = \frac{39+8}{15} \\ = \frac{47}{15}$$

Question 4

$$a) S_n = \frac{a(1-r^n)}{1-r}$$

$$\therefore S_4 = \frac{a(1-r^4)}{1-r} = 30$$

$$S_{\infty} = \frac{a}{1-r} = 32$$

$$\therefore \text{In } S_4 \quad 32(1-r^4) = 30$$

$$1-r^4 = \frac{30}{32}$$

$$r^4 = \frac{2}{32}$$

$$= \frac{1}{16}$$

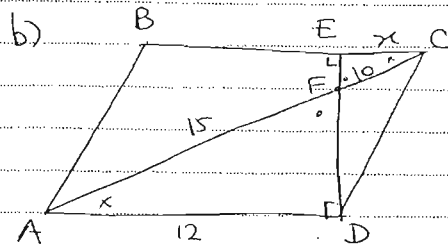
$$r = \pm \frac{1}{2}$$

$$\text{But } r < 0 \quad \therefore r = -\frac{1}{2} \text{ and } a = 48$$

$$\therefore T_1 = 48$$

$$T_2 = -24$$

$$T_3 = 12$$



$$(i) \angle FEC = \angle FDA = 90^\circ \text{ (given)}$$

$$\angle EFC = \angle AFD \text{ (vertically opposite angles equal)}$$

$\therefore \triangle EFC$ and $\triangle DFA$ are equiangular.

$\therefore$ Similar

(ii) Corresponding sides are in the same ratio

$$\therefore \frac{x}{12} = \frac{10}{15}$$

$$x = \frac{10 \times 12}{15}$$

$$= \frac{24}{3}$$

$$= 8$$

$$c) \sin(x + \pi/3) = 0 \quad 0 \leq x \leq \pi$$

$$x + \pi/3 = 0, \pi, 2\pi, 3\pi, \dots$$

$$x = -\pi/3, 2\pi/3, 5\pi/3, 8\pi/3, \dots$$

For given domain:

$$x = 2\pi/3$$

$$d) 2x^2 - 5x + 5 = 0$$

$$(i) \alpha + \beta = 5/2$$

$$(ii) \alpha\beta = 5/2$$

$$(iii) \frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta}$$

$$= \frac{5/2}{5/2}$$

$$= 1$$

Question 5

a) $f(x) = 3x^2 - 2x^3$

(i) $f'(x) = 6x - 6x^2$

for turning points (stationary) $f'(x) = 0$

$\therefore$ Solve $6x(1-x) = 0$

$x = 0, x = 1$

$f''(x) = 6 - 12x$

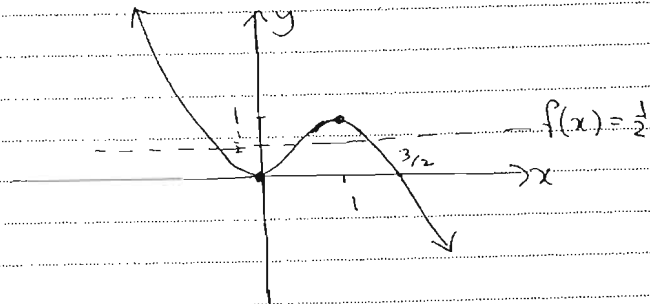
$f''(0) = 6 > 0 \Rightarrow \text{min}$

$f''(1) = 6 - 12 < 0 \Rightarrow \text{max}$

$\therefore$ min at $(0, 0)$

max at $(1, 1)$

(ii)



$f(x) = 0$ when $x^2(3-2x) = 0$
 $\therefore x = 0$ or $x = 3/2$

(iii) $\left. \begin{array}{l} f(x) > 0 \text{ above } y \text{ axis} \\ f'(x) > 0 \text{ increasing} \end{array} \right\} \text{ Both hold for } 0 < x < 1$

(iv) $f(x) = \frac{1}{2}$ (above)

(v) $6x^2 - 4x^3 = 1 \Rightarrow 3x^2 - 2x^3 = \frac{1}{2}$

Since $f(x) = 3x^2 - 2x^3$ and $f(x) = \frac{1}{2}$ intersect 3 times, there will be 3 solutions.

b) $y = -x^2 + 7x - 6$, $y = x + 2$

(i) Intersect when

$-x^2 + 7x - 6 = x + 2$

$\therefore x^2 - 6x + 8 = 0$

$(x-4)(x-2) = 0$

$x = 2$ or $x = 4$

From graph $A = 2$

$B = 4$

(ii) Area = $\int_2^4 (-x^2 + 7x - 6) - (x + 2) dx$

$= \int_2^4 (-x^2 + 6x - 8) dx$

$= \left[-\frac{1}{3}x^3 + 3x^2 - 8x \right]_2^4$

$= -\frac{1}{3}(64) + 3(16) - 32 - \left(-\frac{8}{3} + 12 - 16 \right)$

$= -\frac{64}{3} + 48 - 32 + \frac{8}{3} - 12 + 16$

$= -\frac{56}{3} + 20$

$= \frac{4}{3} \text{ u}^2$

Question 6

$$\begin{aligned} \text{a) } \sum_{r=1}^4 3^{1-r} &= 3^0 + 3^{-1} + 3^{-2} + 3^{-3} \\ &= 1 + \frac{1}{3} + \frac{1}{9} + \frac{1}{27} \\ &= 1\frac{13}{27} \end{aligned}$$

$$\text{b) } 32, 25, 18 \dots \quad a = 32, d = -7$$

$$\begin{aligned} \text{(i) } T_{15} &= a + 14d \\ &= 32 + 14(-7) \\ &= 32 - 98 \\ &= -66 \end{aligned}$$

$$\begin{aligned} \text{(ii) } S_{15} &= \frac{15}{2} [2a + 14d] \\ &= \frac{15}{2} [64 + 14(-7)] \\ &= 15[32 - 49] \\ &= 15 \times -17 \\ &= -255 \end{aligned}$$

$$\begin{aligned} \text{(iii) Sum next 20 terms} &= S_{35} - S_{15} \\ &= \frac{35}{2} [64 + 34(-7)] - (-255) \\ &= 35[32 + 17(-7)] + 255 \\ &= -3045 + 255 \\ &= -2790 \end{aligned}$$

c)

x	0	0.5	1	1.5	2
4^{2x}	1	4	16	64	256
	y_0	y_1	y_2	y_3	y_4

$$\text{Vol} = \pi \int_0^2 4^{2x} dx$$

$$\approx \pi \left[\frac{1}{3} (y_0 + y_4 + 4 \times (y_1 + y_3) + 2(y_2)) \right]$$

$$= \pi \left[\frac{1}{3} (1 + 256 + 4(4 + 64) + 2(16)) \right]$$

$$\text{Vol} = \pi \left[\frac{1}{3} (561) \right]$$

$$= 293.7 \text{ m}^3 \text{ (1 dp)}$$

$$\text{d) (i) } 120^\circ = \frac{2\pi}{3} \text{ rad}$$

$$\begin{aligned} l &= r\theta \\ &= 2 \left(\frac{2\pi}{3} \right) \\ &= \frac{4\pi}{3} \text{ cm} \end{aligned}$$

$$\begin{aligned} \text{(ii) Area} &= \frac{1}{2} r^2 (\theta - \sin \theta) \\ &= \frac{1}{2} (4) \left[\frac{2\pi}{3} - \frac{\sqrt{3}}{2} \right] \\ &= 2 \left(\frac{2\pi}{3} - \frac{\sqrt{3}}{2} \right) \text{ cm}^2 \end{aligned}$$

Question 7

a) $f'(x) = 3x^2 - 4$
 $f(x) = x^3 - 4x + c$

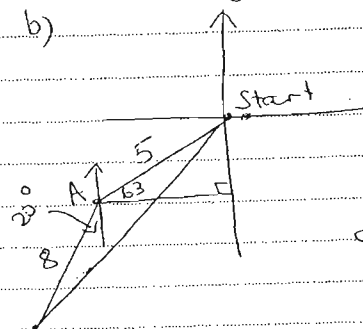
(3, 8) satisfies

$$\therefore 8 = 3^3 - 4(3) + c$$

$$8 = 27 - 12 + c \Rightarrow c = -7$$

$$\therefore y = x^3 - 4x - 7$$

b)



Angle at A = $63 + 90 + 20$
 $= 173^\circ$

$d = \text{distance } S \rightarrow F$

$\therefore$ By cosine rule

$$d^2 = 5^2 + 8^2 - 2 \times 5 \times 8 \cos 173^\circ$$

$$= 25 + 64 - 80 \cos 173^\circ$$

$$= 89 - 80 \cos 173^\circ$$

$$d^2 = 168.4036921$$

$$\therefore d = 12.97704482$$

$$= 13.0 \text{ km (3 sig. figs)}$$

c) \$30000 12% pa = 1% per month
 48 repayments

(i) a) $A_1 = 30000(1.01) - m$

$$b) A_2 = [30000(1.01) - m](1.01) - m$$

$$= 30000(1.01)^2 - m(1.01 + 1)$$

Similarly

$$A_3 = 30000(1.01)^3 - m(1.01^2 + 1.01 + 1)$$

(ii) $A_{48} = 0$ since fully repaid

$$0 = A_{48} = 30000(1.01)^{48} - m(1.01^{47} + 1.01^{46} + \dots + 1)$$

$$\therefore 30000(1.01)^{48} = m(1 + 1.01 + \dots + 1.01^{47})$$

GP with $a=1$ $r=1.1$
 $n=48$

$$\therefore m = \frac{30000(1.01)^{48}}{(1 - 1.01^{48})}$$

$$\frac{1 - 1.01^{48}}{1 - 1.01}$$

$$= \frac{30000(1.01)^{48}(0.01)}{1.01^{48} - 1}$$

$$= \$790.02 \text{ (nearest cent)}$$

(iii) Total repaid = $m \times 48$

$$= \$37920.72 \text{ (nearest cent)}$$

Question 8

$$a) \lim_{x \rightarrow 0} \frac{\sin 2x}{x} = \lim_{x \rightarrow 0} \left(\frac{\sin 2x}{2x} \right) \times 2$$

$$= 2$$

$$b) \log_5 100 - \log_5 4 = \log_5 \left(\frac{100}{4} \right)$$

$$= \log_5 25$$

$$= \log_5 5^2$$

$$= 2$$

$$c) x = 2 + 3t - t^3, \quad t > 0$$

$$(i) \frac{dx}{dt} = 3 - 3t^2$$

$$\text{vel} = 3 - 3t^2$$

$$(ii) \text{ Stops when } v = 0$$

$$\text{i.e. solve } 3 - 3t^2 = 0$$

$$t = 1 \quad (t > 0)$$

Stops after 1 second

$$(iii) t = 0 \text{ in } x = 2 + 3t - t^3$$

$$= 2$$

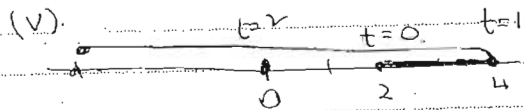
$\therefore$ Initially 2m to the right of O.

$$(iv) \text{ When } t = 2$$

$$v = 3 - 3(2)^2$$

$$= -9$$

$$\text{i.e. } v = -9 \text{ m/sec (travelling to the left)}$$



$$\text{When } t = 1, x = 2 + 3 - 1$$

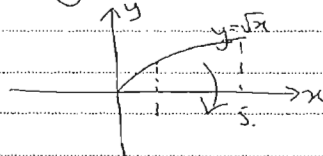
$$= 4$$

$$t = 2, x = 2 + 6 - 8$$

$$= 0$$

$\therefore$ Has travelled $2 + 4 = 6 \text{ m}$

$$d) y = \sqrt{x}$$



$$\text{Vol} = \pi \int_1^5 x \, dx$$

$$= \pi \left[\frac{1}{2} x^2 \right]_1^5$$

$$= \frac{\pi}{2} [25 - 1]$$

$$= 12\pi \text{ u}^3$$

Question 9

a)

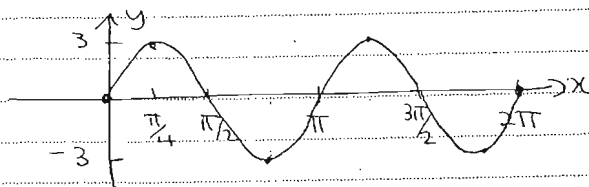
$$f(-1) = (-1)^2 - 2 = -1$$

$$f(1) = 2^1 = 2$$

$$f(10) = \log_{10} 10 = 1$$

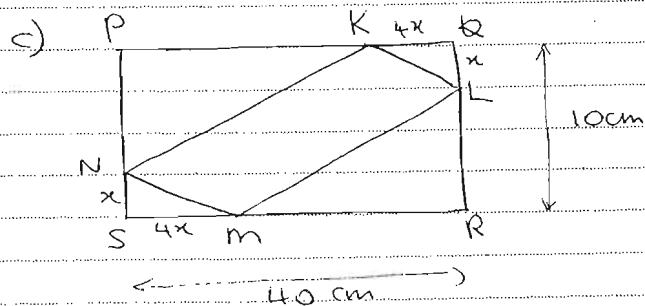
$$\therefore f(-1) + f(1) + f(10) = 2$$

b) $y = 3 \sin 2x$ $0 \leq x \leq 2\pi$



(i) Period = $\frac{2\pi}{2}$
 $= \pi$

(ii) Amplitude = 3



(i) Area parallelogram KLMN

$$= 40 \times 10 - 2 \times \frac{1}{2} (4x)(x)$$

$$- 2 \times \frac{1}{2} (40 - 4x)(10 - x)$$

$$= 400 - 4x^2 - (400 - 40x - 40x + 4x^2)$$

$$= 80x - 8x^2$$

(ii) $0 < x < 10$

(iii) $\frac{dA}{dx} = 80 - 16x$

$$\frac{dA}{dx} = 0 \text{ when } 16x = 80$$

$$x = 5$$

$$\frac{d^2A}{dx^2} = -16 < 0 \Rightarrow \text{max}$$

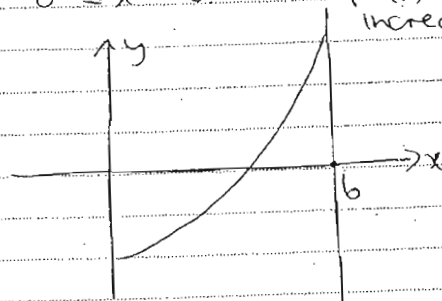
$$\therefore \text{Area max when } x = 5$$

Question 10

a) $0 \leq x \leq 6$

$f'(x) > 0$
increasing

$f''(x) > 0$
concave up.



b) (i) $y = 4 - \sqrt{2x}$

x axis: $y = 0$

i.e. $\sqrt{2x} = 4$

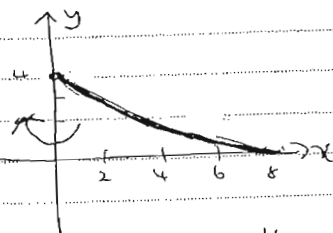
$2x = 16$

$x = 8$

y axis: $x = 0$

i.e. $y = 4$

(ii)



$y = 4 - \sqrt{2x}$

$\sqrt{2x} = 4 - y$

$2x = (4 - y)^2$

$x = \frac{(4 - y)^2}{2}$

$Vol = \pi \int_0^4 x^2 dy$

$= \pi \int_0^4 \frac{(4 - y)^4}{4} dy$

$= \frac{\pi}{4} \left[\frac{(4 - y)^5}{-5} \right]_0^4$

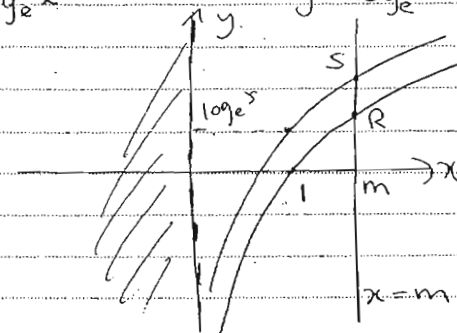
$= -\frac{\pi}{20} [(4 - 4)^5 - (4 - 0)^5]$

$= -\frac{\pi}{20} (-4)^5$

$= \frac{\pi}{20} \times 4^5 = \frac{256\pi}{5}$

c) $y = \log_e x$

$y = \log_e 5 + \log_e x$



$y = \log_e x$

$y' = \frac{1}{x}$

At R, $x = 1$

$\therefore \text{grad} = \frac{1}{1} = 1$

$y = \log_e 5 + \log_e x$

$y' = 0 + \frac{1}{x}$

$= \frac{1}{x}$

At S, $x = 5$

$\therefore \text{grad} = \frac{1}{5}$

$\therefore$ They have the same gradient.

Tangents are parallel.

$R = (1, \log_e 1)$

$S = (5, \log_e 5 + \log_e 5)$

$RS = \sqrt{(5 - 1)^2 + (\log_e 5 + \log_e 5 - \log_e 1)^2}$

$= \sqrt{(4)^2 + (\log_e 5)^2}$

$= \log_e 5$

$\therefore RS$ remains constant

END